

2018 FORMULA STUDENT – ARTIFICIAL INTELLIGENCE (FS-AI) RULES

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INTRODUCTION – FORMULA STUDENT – ARTIFICIAL INTELLIGENCE (FS-AI)

Formula Student UK (FSUK) will hold an event which tests the student's skills to develop software for an Automated Driving System (ADS) that can complete the Dynamic Driving Task (DDT) enabling a Full Size Formula Student Vehicle to complete typical Formula Student Dynamic Events.

Wherever possible, the terminology used in this document follows that defined in SAE J3016 – "Taxonomy and Definitions for Terms Related to Driving Automation Systems". A list of the abbreviations used can be found in the appendix.

As with all Formula Student Competitions, there will be a number of static and dynamic events to test the knowledge and skills of the engineering students.

The Formula Student AI competition will run in parallel with the normal Formula Student event

There will be two classes:

Dynamic Driving Task Class – Formula Student will provide an Automated Driving System Dedicated Vehicle (ADS-DV) which is capable of executing the dynamic driving task when programmed with suitable software. The vehicle is a formula student style vehicle (without driver accommodation) with an electric powertrain, drive-by-wire controls and the required AI (artificial intelligence) computer and sensors for the DDT. The student teams will develop the software for the AI Computer in order that the car may complete its defined missions by executing the Dynamic Driving Task.

Automated Driving System Class – Will follow the Formula Student Driverless technical regulations which allow University teams to use their own Vehicle Platform and adapt that to create their own Automated Driving System - Equipped Vehicle (ADS-EqV).

For teams that intend to compete in Class DDT, for 2018 an ADS-DV will be made available at the competition and it is intended that test days will be held prior to the competition to enable teams to test and develop their software on the actual vehicle.

The static events will cover the design of the ADS software that the teams have produced and the team members understanding of the software design, development and validation. The static events will also include determination of the team's understanding of the design and operational challenges for an ADS to drive in the real world and the requirements for functional safety. As per the main Formula Student event there will also be a business presentation.

For the dynamic events, the ADS vehicle will complete a skidpad mission, an acceleration mission and a 10 lap track drive mission on a special course through the normal Formula Student cones.

A - ADMINISTRATIVE REGULATIONS

A1 ADS AUTOMATED DRIVING SYSTEM (ADS) CLASS INTRODUCTION

ADS Class vehicles must be built to the FSG Driverless Vehicle Rules and follow all FS Driverless regulations from Section A, IC, EV and DV which apply to their vehicle and the Formula Student UK supplementary regulations.

For the static events, the Cost event will be replaced with a Real World Autonomous Event – see rule S4

For the dynamic events, there will be no Efficiency Event as part of the Track Drive.

Points for the ADS Class will be as follows:

Static Events:	
Business Plan Presentation	75
Real World Autonomous	125
Engineering Design	300
Dynamic Events:	
Skid Pad	125
Acceleration	125
Trackdrive	250
Total Points	1000

Table 1: Maximum Points Awarded ADS Class

A2 DYNAMIC DRIVING TASK (DDT) CLASS INTRODUCTION

Formula Student will provide an Autonomous Driving System Dedicated Vehicle (ADS-DV) which is equipped with the following:

- Formula Student style chassis
- Vehicle Control ECU which can receive requests from the Autonomous Computer
- Electric Drivetrain
- Drive-by-wire Steering System
- Drive-by Wire Brakes
- Autonomous Computer – NVIDIA DRIVE PX Parker AutoChauffeur (PX2)
- Sensors including LIDAR, GPS, IMU and Vision cameras

The student teams will develop the Autonomous software which will be compiled and downloaded onto the NVIDIA PX2. This software must enable the following functions:

- Object and Event Detection and Response (OEDR) by interpreting the signals from the provided sensors to determine the location of the cones on the track boundary
- Path planning to determine the route through the cones
- Vehicle position and motion control in both the lateral and longitudinal directions to ensure that the car navigates around the course whilst respecting the track limits by sending steering and wheel torque requests to the vehicle ECU.

Students will have the following tools available to them, however it should be noted that NDAs need to be signed with the respective companies in order to use these tools:

- NVIDIA DriveWorks
- Mathworks Toolchain
- IPG Carmaker

Students are free to use any additional software they wish in order to create their own Software in the Loop (SiL) test environment and to generate code for the NVIDIA PX2 as long as it is compatible with the NVIDIA PX2 when compiled.

DDT Class Points

Static Events:

Business Plan Presentation	75
Real World Autonomous	125
Engineering Design	300

Dynamic Events:

Skid Pad	125
Acceleration	125
Trackdrive	250

Total Points	1000
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Table 2: Maximum Points Awarded DDT Class

For 2018, teams will be allowed to compete in the static class only.

A4 DDT CLASS ADMINISTRATIVE REGULATIONS

A4.9 Autonomous System Responsible (ASR)

A4.9.1 Every participating team in the DDT Class where the vehicle will be operated dynamically must appoint at least one ASR for the competition. This person is responsible for all of the software that is presented to the dynamic events at the competition.

A4.9.4 The ASR must be a valid team member, which means that he/she must have student status, see A4.3.

Note: ADS class entrants must follow all administrative regulations relating to the DV class in the FSG regulations

DV - DDT CLASS TECHNICAL REGULATIONS

This section forms the Technical Rules for DDT Class – Note: The ADS Class must follow FS-DV rules

Note: Wherever possible, the rule number reflects the same rule in the FS-DV regulations.

Introduction

The DDT class will use an ADS-DV which will be provided by the organisers and is equipped with a vehicle ECU and an AI computer and sensors required for the DDT.

No physical modifications to the vehicle will be allowed, however for 2018, the teams may use their own sensors if this is deemed feasible.

The vehicle ECU will be programmed by the vehicle supplier and will directly look after all of the vehicle's actuators (eg. steering and drivetrain). The teams will not be able to make any modifications to the vehicle ECU.

The vehicle will be fitted with an AI computer. This will receive signals from the vehicle sensors either directly or indirectly. The teams will have access to specific areas of the AI computer in order to interpret the sensor data and generate steering and wheel torque or speed targets in order to complete the DDT.

There will be a defined communication protocol between the Vehicle ECU and the AI computer and a software API will be provided to access this in the AI computer.

DV1 VEHICLE INFORMATION

DV 1.3 The vehicle will be fitted with a standardized data logger so that the AI steering and torque targets and other parameters can be monitored. The officials and the team will have access to the recorded data.

DV 1.4 Remote Emergency System (RES) which will be on the DDT Class vehicle.

DV 1.4.2 The ADS-DV vehicle will be equipped with a RES which consists of two parts, a remote control and a vehicle module.

DV 1.4.3 The RES has two functions:

- (a) When the remote emergency stop button is pressed, it will shutdown the ADS-DV in a controlled manner. This is managed by the vehicle ECU and the status will be transmitted to the AI Computer
- (b) Race-control-to-vehicle communication where a "Go" signal is transmitted to the vehicle ECU to give it a "Green Flag" to start.

The "Go" signal will be transmitted to the AI Computer from the ECU

The RES will be operated by the officials.

DV2 AUTONOMOUS SYSTEM

DV2.2 Autonomous System Master Switch (ASMS)

DV2.2.7 The ADS-DV will have an ASMS which may only be switched on after approval from an official. The status of the ASMS will be transmitted to the AI Computer by the vehicle ECU.

DV2.2.8 After switching the ASMS to the "On" position, the vehicle may not start moving and the brakes must remain closed ("AS ready" state, Figure 19) until a "Go" signal is sent via the RES ("AS driving" state, Figure 1).

DV2.4 Autonomous State Definitions – The ADS-DV state will be controlled by the vehicle ECU, but the AI computer will request transitions into the respective active state as appropriate. The expected ADS-DV states as per the FSG regulations are as shown in figure 1. There will be additional states for the systems on the vehicle. The state information will be communicated from the vehicle ECU to the AI computer. The final configuration of the states and communication to the AI computer will be defined in due course.

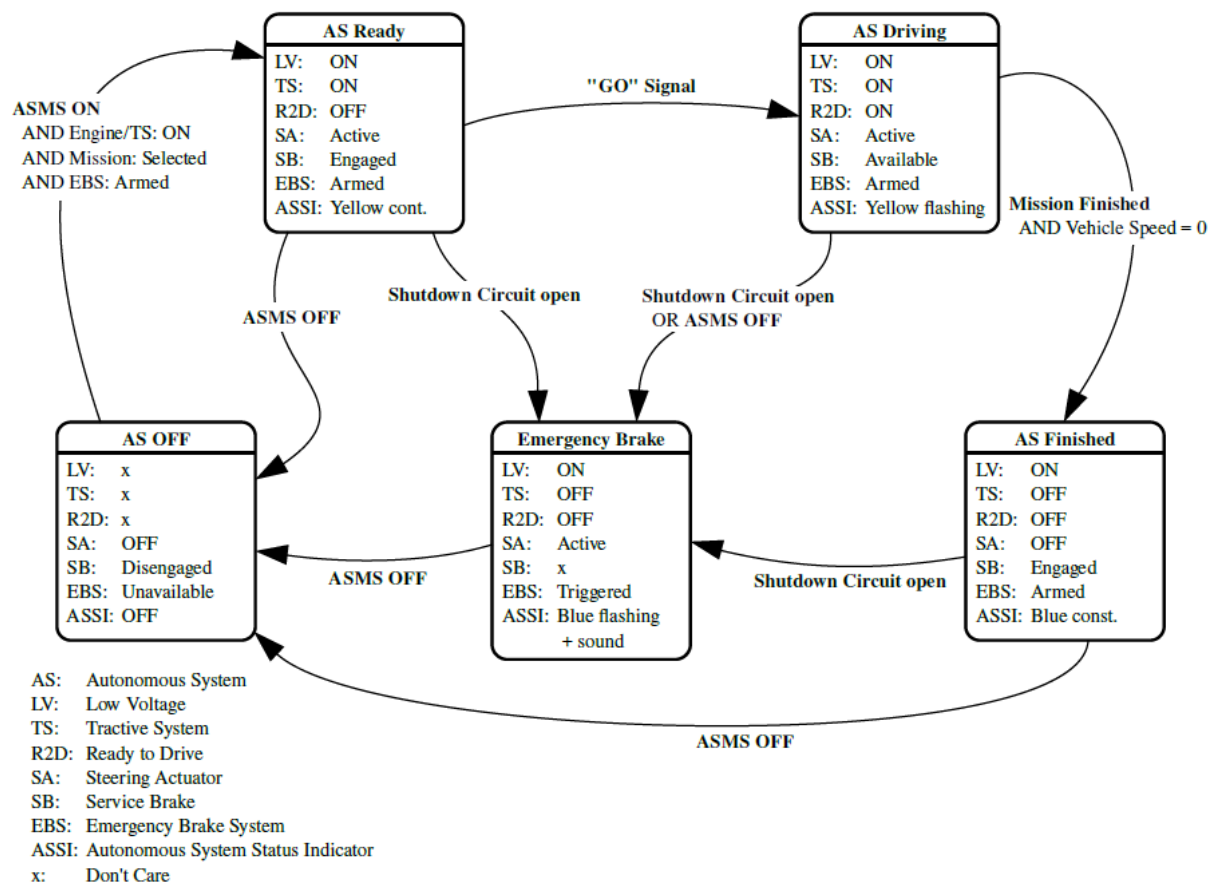


Figure 1 – Autonomous System States

DV2.5 Autonomous Missions

DV2.5.1 The software developed for the AI computer must conduct the Digital Driving Task required to complete the following missions:

- Acceleration
- Skidpad

- Trackdrive
- Autonomous Demo – (IN6)

DV2.5.4 The ADS-DV will have an Autonomous Mission indicator (AMI) which will be checked before every dynamic discipline. The team must ensure that the AI computer sets the AMI state correctly.

DV3 EMERGENCY BRAKING SYSTEM

DV3.2 Functional Safety

DV3.2.1 Requests from AI computer need to be plausible, for example, If the AI computer requests brakes to be applied and the vehicle to accelerate at the same time, then the vehicle ECU will stop the vehicle with the Emergency brake. A list of conditions that are regarded as implausible will be published before the event.

The functional safety of the mechanical systems on the vehicle will be looked after by the vehicle ECU. Any errors will be reported to the AI computer.

DV4 SENSORS, ACTUATORS AND VEHICLE SPEC

Once confirmed, the sensors and actuators will be listed here.

The following sensors will be on the vehicle

- Lidar – Velodyne VLP-16
- 2 x Sekonix AR0231 2MP (SF3323) RCCB GMSL Camera with 120 degrees lens
- PCAN GPS/IMU

The sensor locations will be as follows:

- Lidar – low to the ground at the front of the vehicle
- Cameras – on the shark fin / roll hoop

A full description of the locations is in the document titled “Formula Student ADS-DV Dimensions and Locations.pdf”

The Vehicle specification is as follows:

Drive System:

Drive motors	Saietta 119R-68 (x2)
Peak motor torque	55.7Nm
Peak motor power @ 48V	17.0 kW
Continuous motor torque	27.2 Nm
Continuous motor power @ 48V	8.8 kW
Maximum motor speed	4,000 rpm
Belt drive ratio	3.5:1
Peak motor current	400 A
Motor controllers	Sevcon Gen4 DC (x2)
Operating voltage range	31-90 V
Peak current rating	550 A

Steering System:

Steering Motor	Shiftec ZZ011-1
Steering Controller	Shiftec ZE102-1

Traction Battery:

Cells	CALB CA100AHA
Cell chemistry	Lithium iron phosphate (LiFePO ₄)
Cell capacity rating	100 Ah
Pack configuration	16S1P
Nominal pack energy capacity	5.1 kWh
Nominal pack voltage	51.2 V
Operating voltage range	41.6-58.4 V
Peak discharge current	800 A (up to 10 secs)
Continuous discharge current	300 A
BMS	Proprietary Hypermotive BMS
Charger	TC Charger 3.3 kW (HK-J-H66-40)

DV5 AI COMPUTER AND ECU COMMS

The following commands and signals will be transmitted between the Vehicle ECU and the AI Computer and will be made available through the software API in the PX2:

From Vehicle ECU to AI Computer

- Autonomous System State
- Autonomous Mission Indicator Setting
- Maximum Steering Angle that may be requested (deg)
- Steering Angle (deg)
- 4 x Wheel Speeds (rpm)
- 4 x Wheel Odometry Counts
- 6-axis IMU (3x accelerations and 3x angular rates)

From AI Computer to Vehicle ECU

- Mission Flag (set while mission in progress)
- Steering Angle Request (deg)
- Vehicle Speed Request (wheel rpm)

IN - INSPECTION

IN6 DRIVERLESS INSPECTION (DDT CLASS ONLY)

Note: Class ADS will follow the FS-DV regulations for inspection.

IN6.1 The vehicle must complete an autonomous demo before completing any dynamic events. This demo must sweep the steering through its travel, then with the steering straight, acceleration the vehicle to >20kph and stop the vehicle.

S - STATIC EVENTS

S1 BUSINESS PRESENTATION EVENT

S1.1 Business Presentation Event Objective

S1.1.1 The objective of the presentation event is to evaluate the team's ability to develop and deliver a comprehensive business case that will convince the executives of a corporation that the team's design best meets the needs of the autonomous vehicle development environment, and that it can be profitably manufactured and marketed.

Teams are free to suggest their own manufacturing volume / production rate targets, within their Business Presentations.

S1.1.2 The business presentation event is a role play and teams should present themselves as employees of a company rather than as students. The judges should be treated as if they are executives of a corporation interested in either manufacturing your design or investing in your company. Teams must approach the event with a view to obtaining a business deal to manufacture and sell the team's ADV-DV.

S1.1.3 Teams should assume that the "executives" represent different areas of a corporate organization, including engineering, production, marketing and finance, and thus may not all be engineers.

S1.1.4 Presentations will be evaluated on the contents, organization and visual aids as well as the presenters' delivery and the team's response to questions.

S1.1.5 The presentation must relate to the ADV-DV entered into the competition and the actual quality of the prototype itself will not be considered as part of the presentation judging.

S1.2 Business Presentation Schedule

S1.2.1 The Business Plan Presentation Judging will take place on the static events days. Presentation times will be published in the event schedule. Teams are reminded that the Business Presentations will be held in the Business Presentation judging rooms, NOT at your display area.

S1.3 Business Presentation Format

S1.3.1 One or more team members will give the presentation to the judges.

S1.3.2 All team members who will give any part of the presentation, or who will respond to the judges' questions, must be in the podium area when the presentation starts and must be introduced to the judges. Team members who are part of this "presentation group" may answer the judge's questions even if they did not speak during the presentation itself.

S1.3.3 Presentations are limited to a maximum of ten (10) minutes. Penalties will be imposed if the presentation exceeds 11 minutes or if the presentation is excessively short in duration. Teams will be asked to rapidly conclude their presentation if they overrun significantly.

S1.3.4 The presentation itself will not be interrupted by questions. Immediately following the presentation there will be a question and answer session of up to five (5) minutes.

S1.3.5 Only judges may ask questions. Only team members who are part of the “presentation group” may answer the judges’ questions.

S1.3.6 For the convenience of the Business Presentation Event judges, teams giving a PowerPoint or similar style presentation are required to hand a paper copy of their slides, preferably in colour, to the judges before the presentation begins. A printed copy of the slides is excluded from the supporting material page count defined in S1.3.7. Judges will also arrange for an electronic copy of the presentation to be loaded onto a USB device.

S1.3.7 Content marks will only be awarded for the information presented to the judges. Printed material to support this is allowable but must:

- Be referred to within the presentation.
- Be limited to a maximum of eight (8) sides of A4 paper. Decorative covers are excluded from this page count.

S1.3.8 The format of the final judging will be the same as for the initial judging including Questions & Answers. The “presentation group” for the final must consist of the same team members as the initial judging.

S1.3.9 Access to an internet connection during the initial or final judging cannot be guaranteed.

S1.4 Data Projection Equipment

S1.4.1 LCD / Plasma TV-style screens or video projectors will be provided by the organisers, but teams should bring their own laptop computers and may use their own projectors if they wish. The screens/projectors will have HDMI and VGA Input Connectors. Teams are responsible for the compatibility of their computer equipment and setting up of the screens. Overseas teams should ensure they have UK compatible power leads/adaptors.

S1.5 Evaluation Criteria

S1.5.1 Forty (40) % of the marks are allocated to Content; the remaining marks are equally allocated to Organisation, Visual Aids, Delivery and Q&A. The broad topics are in line with the standard FSAE scoring sheet.

S1.5.2 The criteria are applied only to the team’s presentation itself. The team that makes the best presentation, regardless of the quality of their car, will win the event.

S1.6 Scoring Formula

S1.6.1 The maximum points from the Business Presentation event is 75

S1.6.2 The Business Presentation scores will be normalised such that the highest scoring presentation scores 75 points towards the overall competition score, all other teams will be awarded points on a pro-rata basis.

BUSINESS PRESENTATION SCORE = $(75) \times P_{\text{your}}/P_{\text{max}}$

Where:

- “Pmax” is the highest score awarded to any team
- “Pyour” is the score awarded to your team

S3 STATIC DESIGN EVENT CLASS ADS AND DDT

S3.1.1 Class ADS will present their vehicle design at the vehicle and will be judged in the same manner as a Class 1 entry.

S3.1.2 Class DDT will have a presentation room with LCD display available. They may present a 10 minute presentation at the start of the event or present posters of their software design and simulation. There will be a further 20 minutes to discuss the content of their presentation, the design of their software, simulation, integration and topics such as functional safety as required to develop the software for an autonomous vehicle.

S3.4 Autonomous Design Report (ADR) – Note: the ADR is optional for class DDT

S3.4.2 The ADR should contain a description of the autonomous system with a review and derivation of the team's design objectives. Any information to scope, explain or highlight design features, concepts, methods or objectives to express the value and performance of the autonomous system to the judges shall be included at the team's discretion.

S3.4.3 Evidence of information mentioned in the ADR should be brought to the competition and be available, on request, for review by the judges.

S3.4.4 The ADR must not exceed five pages of content (text, which may include pictures and graphs).

S3.4.5 Any portions of the ADR that exceed five pages of content will not be evaluated.

S3.4.6 The ADR must be written as a scientific paper.

S3.5.5 Teams need to show some simulation test data, where the judges can see how the algorithms work and access to the software and algorithms must be provided.

ENGINEERING DESIGN SCORE = $(300) \times P_{\text{your}}/P_{\text{max}}$

Where:

- "Pmax" is the highest score awarded to any team
- "Pyour" is the score awarded to your team

S4 REAL WORLD AUTONOMOUS EVENT

S4.1 Real World Autonomous Event Objective

S4.1.1 The objective of the real world autonomous event is to evaluate the team's understanding of the challenges to be overcome in order that an autonomous vehicle may operate in the real world. This will include interactions with the infrastructure (roads, road markings, traffic lights and signposts), other road users (cars, trucks and buses and cyclists etc) and interactions with pedestrians and other obstacles. Functional safety for autonomous vehicles and the considerations for the real world must also be considered in detail.

S4.1.2 The real world autonomous presentation event is predominantly a technical event and will focus primarily on the team's understanding of the real world challenging for Autonomous vehicles.

S4.1.3 The team should focus on the challenges that would be faced for their ADS vehicle to meet SAE level 4 and can additionally comment on the additional challenges to achieve level 5.

S4.1.4 Presentations will be evaluated on the contents, organization and visual aids as well as the presenters' delivery and the team's overall knowledge and response to questions.

S4.2 Real World Autonomous Presentation Schedule

S4.2.1 The Real World Autonomous Judging will take place on the static events days. Presentation times will be published in the event schedule. Teams are reminded that the Real World Autonomous Presentations will be held in the Real World Autonomous Presentation judging rooms, NOT at your display area.

S4.3 Real World Autonomous Presentation Format

S4.3.1 One or more team members will give the presentation to the judges.

S4.3.2 All team members who will give any part of the presentation, or who will respond to the judges' questions, must be in the podium area when the presentation starts and must be introduced to the judges. Team members who are part of this "presentation group" may answer the judge's questions even if they did not speak during the presentation itself.

S4.3.3 Presentations are limited to a maximum of fifteen (10) minutes. Penalties will be imposed if the presentation exceeds 11 minutes or if the presentation is excessively short in duration. Teams will be asked to rapidly conclude their presentation if they overrun significantly.

S4.3.4 The presentation itself will not be interrupted by questions. Immediately following the presentation there will be a question and answer session of up to ten (15) minutes.

S4.3.5 Only judges may ask questions. Only team members who are part of the "presentation group" may answer the judges' questions.

S4.3.6 For the convenience of the Real World Autonomous Presentation Event judges, teams giving a PowerPoint or similar style presentation are required to hand a paper copy of their slides, preferably in colour, to the judges before the presentation begins. A printed copy of the slides is excluded from the supporting material page count defined in S4.3.7. Judges will also arrange for an electronic copy of the presentation to be loaded onto a USB device.

S4.3.7 Content marks will only be awarded for the information presented to the judges. Printed material to support this is allowable but must:

- Be referred to within the presentation.
- Be limited to a maximum of eight (12) sides of A4 paper. Decorative covers are excluded from this page count.

S4.3.8 There will be no final for the Real World Autonomous Event.

S4.3.9 Access to an internet connection during the initial or final judging cannot be guaranteed.

S4.4 Data Projection Equipment

S4.4.1 LCD / Plasma TV-style screens or video projectors will be provided by the organisers, but teams should bring their own laptop computers and may use their own projectors if they wish. The screens/projectors will have HDMI and VGA Input Connectors. Teams are responsible for the compatibility of their computer equipment and setting up of the screens. Overseas teams should ensure they have UK compatible power leads/adaptors.

S4.5 Evaluation Criteria

S4.5.1 Forty (40) % of the marks are allocated to Content, Twenty (20) % of the marks are allocated to Organisation, Visual Aids, Delivery and Forty (40) % of the marks are allocated to the Q&A session.

S4.5.2 The criteria are applied to the team's presentation and the team's understanding of the subject material.

S4.6 Scoring Formula

S4.6.1 The maximum points from the Real World Autonomous Presentation event is 125

S4.6.2 The Real World Autonomous Presentation scores will be normalised such that the highest scoring presentation scores 125 points towards the overall competition score, all other teams will be awarded points on a pro-rata basis.

REAL WORLD AUTONOMOUS PRESENTATION SCORE = $(125) \times P_{\text{your}}/P_{\text{max}}$

Where:

- "Pmax" is the highest score awarded to any team
- "Pyour" is the score awarded to your team

D - DYNAMIC EVENT REGULATIONS

D2 AUTONOMOUS DRIVING RULES

D2.2.8 When driving autonomously, an ASR has to be present at the race control with the RES. Additionally, one single monitoring device (laptop, tablet, ...) may be brought.

D2.4.2 A practice track for DV will be available for testing in autonomous mode.

D2.5 Cones & Markings

D2.5.1 Details of the cones used and more detailed track layout figures will be provided shortly.

D2.6 Start-up Procedure – There will be a time limit for the run to start after the “Go” signal is sent.

D2.6.1 No additional equipment (e.g. laptop, jack-up device, pressure tank, etc.) is allowed to start up the vehicle at the staging/starting line.

D2.6.2 If the vehicle does not enter “AS Ready” state within 1 min after being staged, the team may be sent back to the preparation area by the officials.

D2.6.3 The vehicle may only be staged with the steering system in straight position.

D2.6.4 The vehicle may be pushed from the preparation area to the start line with activated GLVS

D2.6.5 The EBS may be armed already in the preparation area.

D2.7 Vehicle Break Downs and Usage of RES – Class ADS

D2.7.1 Stalling the engine or deactivating the tractive system for any reason during a dynamic event will result in DNF as the autonomous system is not allowed to restart the engine/reactivate the tractive system.

D2.7.2 If a vehicle comes to standstill for any reason, it may have up to 30 s to attempt to continue to drive. If the vehicle doesn’t restart within 30 s, it will be deactivated using the RES, deemed disabled and scored as DNF for the run.

D2.7.3 The ASR or the officials may stop the vehicle using the RES in any of the following cases:

- Its behaviour seems to be uncontrolled (e.g. driving off-course).
- It is mechanically or electrically damaged.
- It is driving too slowly.
- To ensure safe conditions on the track (e.g. persons or animals on the track). In this case the team will get a re-run.

D2.7.4 If a vehicle breaks down or is stopped by the use of the RES it will be removed from the course, will not be allowed to re-enter the course and scored DNF.

D2.7.5 If a traceable signal loss of the RES appears and doubtless proof can be brought by the team that it is was not self-inflicted, a re-run may be granted.

D2.7.6 At the direction of the officials, team members may be instructed to retrieve broken-down vehicles. This recovery may only be done under the control of the officials.

D2.8 Vehicle Break Downs and Usage of RES – Class DDT

D2.8.1 Deactivating the tractive system by the AI computer during a dynamic event will result in DNF as the autonomous system is not allowed to reactivate the tractive system.

D2.8.2 If a vehicle comes to standstill for any reason, it may have up to 30 s to attempt to continue to drive. If the vehicle doesn't restart within 30 s, it will be deactivated using the RES, deemed disabled and scored as DNF for the run unless it is a vehicle issue which is outside of the control of the team.

D2.8.3 The officials may stop the vehicle using the RES in any of the following cases:

- Its behaviour seems to be uncontrolled (e.g. driving off-course).
- It is mechanically or electrically damaged.
- It is driving too slowly.
- To ensure safe conditions on the track (e.g. persons or animals on the track). In this case the team will get a re-run.

D2.8.4a If the vehicle breaks down a re-run will be allowed.

D2.8.4b If the vehicle is stopped by the use of the RES it will be removed from the course, will not be allowed to re-enter the course and scored DNF.

D2.8.5 If a traceable signal loss of the RES appears and doubtless proof can be brought by the team that it is was not self-inflicted, a re-run may be granted.

D2.8.6 The officials will deal with recovery of the vehicle.

D2.9 Procedure After Completing a Dynamic Event – Class ADS

D2.9.1 Class ADS vehicles must be collected by the ASR and an additional team member immediately after approval from the officials.

D4 SKIDPAD EVENT

D4.3 Skidpad Procedure

D4.3.1 Each team has at least two runs. The final number of runs will be published before the start of the event.

D4.3.2 Starting order is based upon time of arrival. Teams on their first run will receive priority.

D4.3.3 Staging - The foremost part of the vehicle is staged 15m in front of the timekeeping line.

D4.3.4 Starting - A go-signal from RES is used to indicate the approval to begin.

D4.3.5 The vehicle will enter perpendicular to the figure eight and will take one full lap on the right circle to establish the turn. The next lap will be on the right circle and will be timed.

Immediately following the second lap, the vehicle will enter the left circle for the third lap.

The fourth lap will be on the left circle and will be timed. Immediately upon finishing the fourth lap, the vehicle will exit the track.

D4.3.6 The vehicle will exit at the intersection moving in the same direction as entered and must come to a full stop within 25m behind the timekeeping line, inside the marked exit lane and enter the finish-state described in DV2.3.

D4.4 Skidpad Scoring

D4.4.4 If a team's run time including penalties is below T_{max} , additional points based on the following formula are given:

D4.4.5 25 points are awarded to every team that finishes at least one run without a DNF.

D4.4.6 If a team's run time including penalties is below T_{max} , additional points based on the following formula are given:

$$SKIDPADSCORE = 125 \left(\frac{\left(\frac{T_{max}}{T_{team}} \right)^2 - 1}{\left(\frac{T_{max}}{T_{min}} \right)^2 - 1} \right)$$

T_{team} is the team's best run time including penalties.

T_{min} is the fastest vehicle run time including penalties.

T_{max} is 200% of the fastest vehicle time including penalties.

D5 ACCELERATION EVENT

D5.1.2 The course is 3m wide.

D5.3 Acceleration Procedure

D5.3.1 Each team has at least two runs. The final number of runs will be published before the start of the event.

D5.3.2 Staging - The foremost part of the vehicle is staged at 0.30m before the starting line. Vehicles will accelerate from a standing start.

D5.3.3 Starting - A go-signal from RES is used to indicate the approval to begin, timing starts only after the vehicle crosses the starting line and stops after it crosses the finish line.

D5.3.4 After the finish line, the vehicle must come to a full stop within 100m inside the marked exit lane and enter the finish-state described in DV2.3.

D5.3.5 Starting order is based upon time of arrival. Teams on their first run will receive priority.

D5.4 Acceleration Scoring

D5.4.1 25 points are awarded to every team that finishes at least one run without a DNF.

D5.4.2 If a team's best time including penalties is below $T_{\max}$ additional points based on the following formula are given:

$$ACCELERATION\ SCORE = 125 \left(\frac{\frac{T_{\max}}{T_{team}} - 1}{\frac{T_{\max}}{T_{min}} - 1} \right)$$

T_{team} is the team's best time including penalties.

T_{min} is the fastest vehicle time.

$T_{\max}$ is 200% of the fastest vehicle time.

D8 TRACKDRIVE

D8.1 Trackdrive Track layout

D8.1.1 The trackdrive layout is a closed loop circuit built to the following guidelines:

- Straights: No longer than 80m
- Constant Turns: up to 50m diameter
- Hairpin Turns: Minimum of 9m outside diameter (of the turn)
- Miscellaneous: Chicanes, multiple turns, decreasing radius turns, etc.
- The minimum track width is 3m

D8.1.2 The length of one lap is approximately 200m to 500m.

D8.2 Trackdrive Procedure

D8.2.1 Starting order will be defined by the officials, based on acceleration and skidpad.

D8.2.2 There is no requirement for the fuel level to be set before the event.

D8.2.3 There will be at least one run consisting of ten laps. The number of runs and the starting order procedure will be announced before the start of the event.

D8.2.4 Staging - The vehicle is staged such that the front wheels are 6m in before the starting line on the track.

D8.2.5 Starting - A go-signal from RES is used to indicate the approval to begin. Timing starts after the vehicle crosses the starting line.

D8.2.6 After ten laps the vehicle must come to a full stop within 30m behind the finish line on the track and enter the finish-state described in DV2.4.

D8.2.7 There will be no last lap signal i.e. the vehicle should count laps itself.

D8.3 Trackdrive Course Walk

D8.3.1 There will be a course walk prior to the trackdrive. During the course walk no equipment (e.g. antennas, sensors, cameras, GPS, etc) other than analog measurement devices (i.e. measurement wheel or measurement tape) is allowed.

D8.4 Trackdrive Scoring

D8.4.1 Each lap of the trackdrive event is individually timed. The corrected elapsed time is determined by adding any penalty times.

D8.4.2 Ten points are awarded for every completed lap, independent of the corrected elapsed time. This is also applied for teams that do not finish the trackdrive i.e. get a DNF.

D8.4.2 If a team's best time including penalties is below $T_{\max}$ additional points based on the following formula are given:

$$TRACKDRIVE\ SCORE = 250 \left(\frac{\frac{T_{\max}}{T_{team}} - 1}{\frac{T_{\max}}{T_{min}} - 1} \right)$$

T_{team} is the team's best time including penalties.

T_{min} is the fastest vehicle time.

$T_{\max}$ is 200% of the fastest vehicle time.

T_{team} is the team's corrected elapsed time.

T_{max} is 2 times of the corrected elapsed time of the fastest vehicle.

D9.1.3 Cones that are DOO are not replaced/reset during the run. There will be no re-run due to cones in the driving path or disorientation due to missing cones.

D9.1.6 An Unsafe Stop (USS) is defined as not stopping within the specified area and/or not entering the finish-state described in DV2.4.

APPENDIX - ABBREVIATIONS AND NOMENCLATURE

ADR	Autonomous Design Report
ADS	Automated Driving System
ADS-DV	Automated Driving System Dedicated Vehicle
ADS-EqV	Automated Driving System Equipped Vehicle
AI Computer	Artificial Intelligence Computer
AMI	Autonomous Mission Indicator
ASMS	Autonomous System Master Switch
ASR	Autonomous System Responsible
DDT	Dynamic Driving Task
DOO	Down or Out (Cones)
DV	Driverless Vehicle (From FSG competition regulations)
ECU	Electronic Control Unit
HiL	Hardware in the Loop
Missions	Dynamic Driving Tasks that the vehicle must complete
OEDR	Object and Event Detection and Response
RES	Remote Emergency System
USS	Unsafe Stop